

Holberton Portfolio Project - Team Formation and Idea Development (Stage 1)

Objectives:

- Team Formation
- Research and Brainstorming
- Idea Evaluation
- Decision and Refinement
- Documentation

Deliverables:

- Selected project idea
- MVP Overview
- Brainstorming and analysis

Introduction:

This project is the final deliverable of the Holberton Backend Programming course. The focus of it is to simulate the developing environment and planning of a professional software product. The first stage of this project is to gather the team members, brainstorm and evaluate project ideas.

Team members:

- Tito Alberto Mundo Rodríguez

Research and Brainstorming:

Brainstormed ideas:

- **QR/Code Reservation Manager for residencies/businesses:** Services to generate access codes for reservations and security access. Works via numbercodes and maybe QR Codes.
- **Inventory Manager:** Platform to manage storage and inventory for stores and businesses, manage different types of items, quantities and warehouses.

- **Cloud Storage:** Web platform to upload and store files and documents. Works like Google Cloud or DropBox. Focused more on rolebase access to files.

Idea	Pros	Cons
Reservation Manager	- Low technical difficulty. - Easy to use. - Big market share.	- Maybe it would work better as a mobile app. - Having to manage QR generation.
Inventory Manager	- Simple. - Low technical difficulty.	- Manage items between warehouses. - Manage different types of items and units.
Cloud Storage	- Big market share and opportunities. - Rolebase access to files.	- High technical difficulty. - Needing an external service like AWS S3 for storage.

“How could we give users access to the reservations?” We could generate a simple 6-8 digit number. With 8 digits we would have 10 million different codes for each place. We could also generate QR Codes with the UUID of the reservation for ease of use. QR codes would be convenient for mobile users and numerical codes would be easier for desktop users.

“How could we get and afford the storage for the Cloud Storage service?” We can use a service like Amazon S3 buckets to have scalable storage. We do not need to worry about managing our own file storage and S3 costing scales with usage, so it is not very expensive.

“How could you manage different types of items for the Storage/Inventory Manage?” We need to give the ability to users to create their own items and store the qualities of that kind of item in a database.

Idea Evaluation:

We use the following table to define the feasibility, potential impact, technical alignment and scalability of each different idea rated from 1 to 5. (1 being the lowest and 5 being the highest).

Idea	Feasibility	Potential Impact	Technical Alignment	Scalability
Reservation Manager	4/5	4/5	3.5/5	4.5/5
Inventory Manager	4.5/5	4/5	4/5	4.5/5
Cloud Storage	4/5	4.5/5	3/5	3.5/5

Overall the simplest to implement is the Inventory Manager, it doesn't need any specific technical requirement unlike the other two options, but would benefit from them. The reservations manager with codes would work fine but a mobile app would be more convenient due to the ability to use QR codes. Lastly the Cloud Storage option is interesting but also very difficult due to being technically complex and needing more infrastructure due to the storage necessity.

Decision and Refinement:

Selected Idea: Reservation Manager.

We decided to choose the Reservation Manager to strike a balance between something simple that still gets a more specific function and does not require a specific infrastructure.

Final concept: Bookworm.

Problem Solved: Restaurants, offices or businesses in need of a system for booking reservations. Allowing users to book, create, manage and close the reservations.

Specific challenges to handle:

- Management of dates and overlapping reservations.
- Separate regular user and admin roles (Role based views).
- See pending, ongoing and finished reservations.

Required functions:

- Create reservations/turns.
- Book/schedule reservations.
- Activated booked reservations.
- Modify and delete reservations.
- See pending, current and finished reservations.